

Quality Management

SE1001 -Intro to Software Engineering

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Topics covered

- Software quality
- Quality attributes

Quality Concepts

Can anybody define quality ?

**Which one is better quality product
Suzuki or Mercedes?**



What is Quality?

- Quality, simplistically, means that a product should meet its specification.
- This is problematical for software systems
 - There is a tension between customer quality requirements (efficiency, reliability, etc.) and developer quality requirements (maintainability, reusability, etc.);
 - Some quality requirements are difficult to specify in an unambiguous way;
 - Software specifications are usually incomplete and often inconsistent.

The Quality Compromise

- We cannot wait for specifications to improve before paying attention to quality management.
- We must put quality management procedures into place to improve quality in spite of imperfect specification.

Software Quality Management

- Concerned with ensuring that the required level of quality is achieved in a software product.
- Involves defining appropriate quality standards and procedures and ensuring that these are followed.
- Should aim to develop a 'quality culture' where quality is seen as everyone's responsibility.

Software Quality Management

- **Two principal concerns:**
 - At the organizational level, quality management is concerned with establishing a framework of organizational processes and standards that will lead to high-quality software.
 - At the project level, quality management involves the application of specific quality processes and checking that these planned processes have been followed. Project quality management is also concerned with establishing a quality plan for a project. The quality plan should set out the **quality goals** for the project and define what processes and standards are to be used.

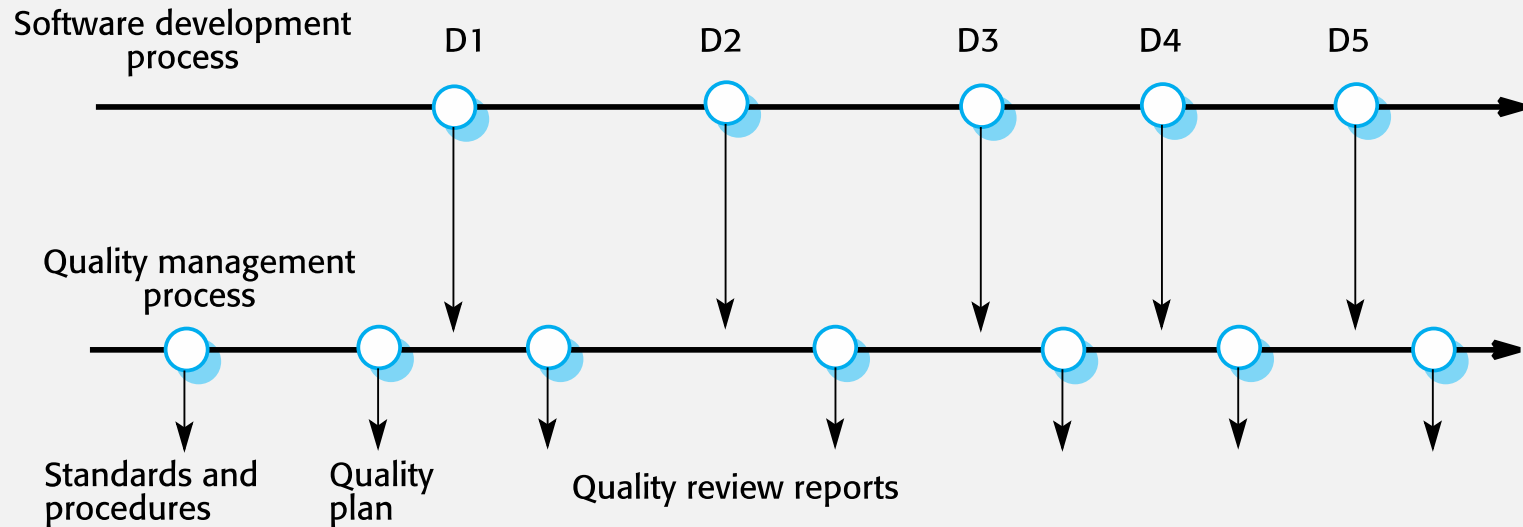
Quality Management Activities

- Quality management provides an independent check on the software development process.
- The quality management process checks the project deliverables to ensure that they are consistent with organizational standards and goals.
- The quality team should be independent from the development team so that they can take an objective view of the software. This allows them to report on software quality without being influenced by software development issues.

Quality Management Activities

- Quality assurance
 - Establish organisational procedures and standards for quality.
- Quality planning
 - Select applicable procedures and standards for a particular project and modify these as required.
- Quality control
 - Ensure that procedures and standards are followed by the software development team.
- Quality management should be separate from project management to ensure independence.

Quality Management and Software Development



Quality Planning

- A quality plan sets out the **desired product qualities** and how these are assessed and defines the most significant **quality attributes**.
- The quality plan should define the **quality assessment process**.
- It should set out which **organisational standards** should be applied and, where necessary, define new standards to be used.

Non-functional Characteristics

- The subjective quality of a software system is largely based on its non-functional characteristics.
- This reflects practical user experience – if the software's functionality is not what is expected, then users will often just work around this and find other ways to do what they want to do.
- However, if the software is unreliable or too slow, then it is practically impossible for them to achieve their goals.

Software Quality Attributes

Safety	Availability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

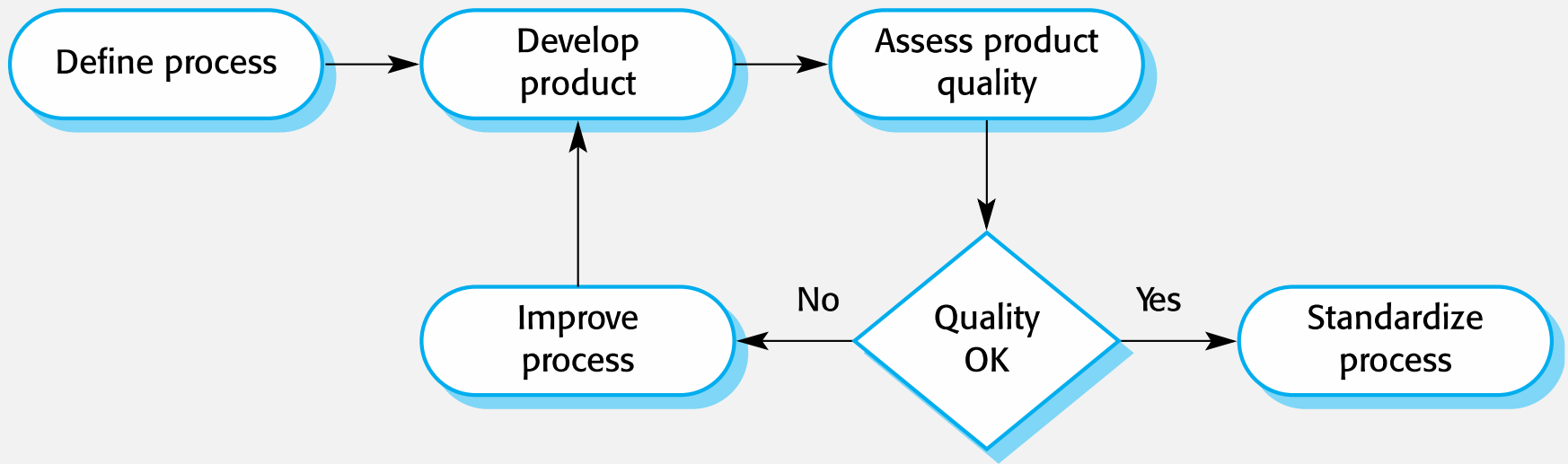
Quality Conflicts

- It is not possible for any system to be optimized for all of these attributes – for example, improving security may lead to loss of performance.
- The quality plan should therefore **define the most important quality attributes** for the software that is being developed.
- The plan should also include a definition of the **quality assessment process**, an agreed way of assessing whether some quality, such as maintainability or robustness, is present in the product.

Process and Product Quality

- The quality of a developed product is influenced by the quality of the production process.
- This is important in software development as some product quality attributes are hard to assess.
- However, there is a very complex and poorly understood relationship between software processes and product quality.
 - The application of individual skills and experience is particularly important in software development;
 - External factors such as the novelty of an application or the need for an accelerated development schedule may impair product quality.

Process-based Quality





That is all